

Kundai Farai Sachikonye

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🌐 github.com/fullscreen-triangle



Bioinformatician: NGS and omics pipelines, sequence and variant analysis, and database development — reproducible, validated, and at scale.

Profile

Bioinformatician building NGS and omics analysis pipelines, sequence and variant analysis, and biological databases. I deliver reproducible, HPC-scale workflows validated against authoritative reference data.

Technical Skills

Sequence analysis	Variant calling, RNA-seq, transcriptome analysis; multi-omics integration
Mass spectrometry	Proteomics / metabolomics pipelines; metabolite identification
Pipelines & databases	Reproducible workflows; relational database design (lipidomics databases); HDF5, Parquet
HPC & cloud	Ray, Dask, SLURM; AWS; Docker
Programming	Python (primary), R, Bash, SQL, Rust
Reference data	1000 Genomes, gnomAD, UK Biobank, IEDB, NIST

Experience

2021–present	Independent Bioinformatics Researcher, Bitspark GmbH Omics pipelines, bioinformatics tools, and biological databases; deployed, documented, reference-validated.
2019–2021	Computational Lipidomics — Bioinformatics & HPC, Technische Universität München, Munich High-throughput mass-spectrometry bioinformatics at HPC scale; >100 GB datasets. Taught at Master's level.
2017–2018	Glycomics & Proteomics, Universitätsklinikum Hamburg-Eppendorf, Hamburg Automated LC-MS/MS and MALDI-TOF data pipelines.
2013	Cancer Biomarker Discovery, University Medical Center Groningen, Groningen UPLC-MS/MS shotgun proteomics; high-dimensional data analysis.

Selected Projects

gospel	Multi-omics integration pipeline; validated against 1000 Genomes, gnomAD, UK Biobank. github.com/fullscreen-triangle/gospel
lavoisier	High-throughput mass-spectrometry pipeline; 98.9% NIST accuracy; distributed (Ray). lavoisier.vercel.app/sandbox

Lipidomics Designed and built relational databases for lipidomics data — schema, ingestion, and query.
databases

syndrome Immunopectidomics — neoantigen / MHC prediction (AUC 0.81, IEDB). syndrome-
seven.vercel.app

Education

2018–2019 **MSc Computational Biology**, *Universität Tübingen*

2014–2017 **MSc Pharmaceutical Biotechnology**, *HAW Hamburg*

2011–2014 **BSc Biochemistry and Cell Biology**, *Jacobs University Bremen*

Additional: doctoral research, computational lipidomics, TUM (not completed).

Languages

English Native / Fluent (C2)

German Conversational

resident in Germany since 2011